

MNIST and EEG signal visualization and classification through dimension reduction

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1 Abstract

Observing and visualizing a graph is, in most cases, not too difficult. Lots of different tools already exists and in the case of pictures (a very common type of graph indeed), the visualization is obvious and actually makes sense. However, visualizing and comparing different graphs can be hard, especially with large graphs or large numbers of graphs, something that happens commonly in, say, machine learning. Graph visualization can find its utility in helping us getting a better understanding of the properties of the graphs.

2 Choice of subject and objectives

The first set of graphs we choose was the MNIST database[1]. It's a classic choice, motivated by the size (60000 pictures) and the fame of the MNIST database. Then, we chose a database of electroencephalography (EEG) tests, consisting in people grasping and lifting different weight (each participant had to move their hand to grasp an object, to lift and replace it in under 10 seconds, with a sampling rate of 500Hz)[2]. Both can be understood as graph, with each node being a pixel or an electrode. In the MNIST dataset, two nodes are connected with an edge if they are adjacent in the picture. In the EEG dataset, the underlying graph is made by considering the shape of the helmet and the connections of the electrodes (figure 1).

The differences between the dataset lie in the

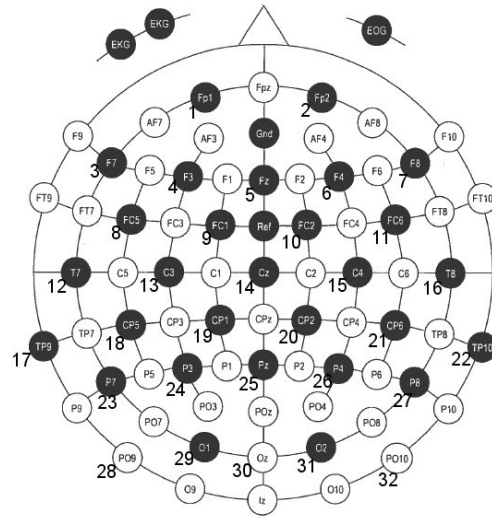


FIGURE 1 – EEG graph shape

shape of each set : MNIST's graph are made of 784 nodes disposed in a grid shape where EEG ones are made of 32 nodes in a pseudo-grid shape. Each graph of the MNIST dataset corresponds to a picture of a digit(the represented digit being the label), with the value on each node being the value of each pixel. With the EEG dataset, each graph corresponds to a moment (the performed action being the label), where the value on each node is the value of the corresponding electrode.

The goal of this study is to experiment and to see how the same methods result in different ways with more or less efficiency. The ideal result would be easy-to-understand and meaningful graphs.

3 Used Methods

3.1 t-distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is an algorithm used for dimensionality reduction. It is best used on high-dimensional data transformed into a two or three dimensions space; therefore, this algorithm is well suited for 2D or 3D visualization of high-dimensional data. The reduction is made by computing on the similarities between the points, making it more effective in dataset with well-defined classes.[3]

3.2 k-Nearest Neighbors Algorithm

The k-Nearest Neighbors algorithm (or k-NN for short) functions by classifying an object by observing the k-nearest neighbors and deciding the type of the object according to the majority. For instance, with $k=1$, every new object will be recognized as its nearest neighbor. For k-NN with $k>1$, the weight of each neighbor can be pondered with its distance to the object; a common choice is $1/d$, with d as the distance to the neighbor. Finally, the k-NN algorithm can be used in One-vs.-rest case to improve the result by making two classes : one containing the class we want to identify and the other class containing every other classes. An less specific case is to simply part the data in two parts that we want to differentiate.

3.3 Supervised, unsupervised and semi-supervised learning

In machine learning, supervised learning occurs when the training is done with fully-labeled data. The result is usually the most satisfying; however, it requires that the whole dataset is correctly labeled.

Opposed to it is the unsupervised learning. It occurs when the data are entirely not labeled. Different algorithms can be used to regroup the

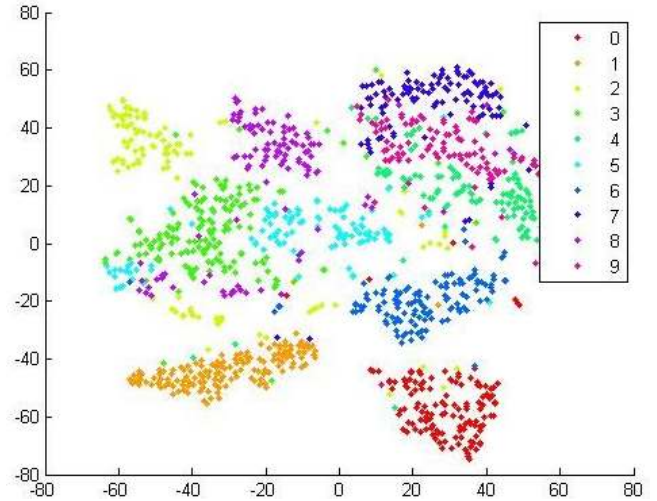


FIGURE 2 – 2D repartition of the MNIST data after t-SNE

data in different classes, the choice of the algorithm depending on the background knowledge of the data sample.

Then, semi-supervised occurs when the data are partially labeled. The training first begins with the labeled data, then adds the unlabeled data to the training sample.

4 MNIST Results

4.1 Visualization

We first use the t-SNE algorithm to reduce the dimension of the dataset down to two dimensions, then we represent them on a scatter plot (figure 2).

For the figure 1, we chose randomly 1200 digits for data size limitation. The first dimension resulting of t-SNE is assigned to X and the second dimension is assigned to Y. We can see on the plot that most classes contains elements that share similarities in two dimensions, thus they are also similar in 784 dimensions. For comparison's and demonstration's stake, a reduction in three dimensions is shown in (figure 3).

Then, a 0 versus the rest example shows the clear separation between a group and the rest(figure 4).

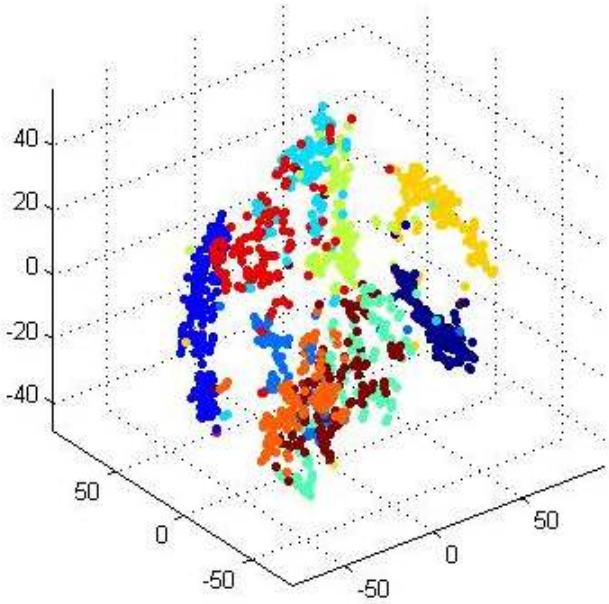


FIGURE 3 – 3D repartition of the MNIST data after t-SNE

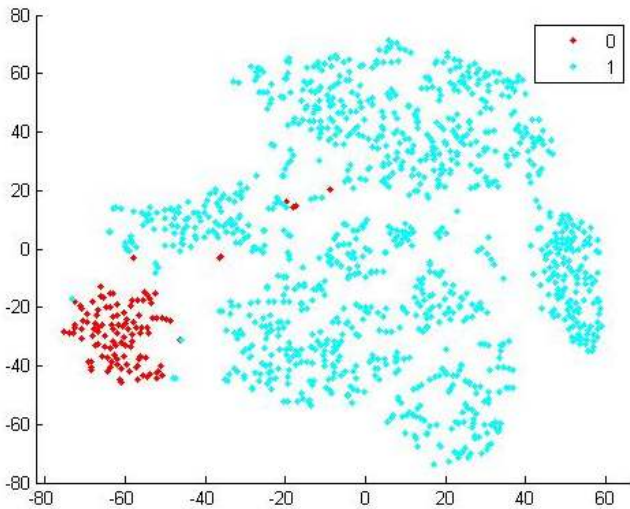


FIGURE 4 – Example of 0 versus Rest in MNIST

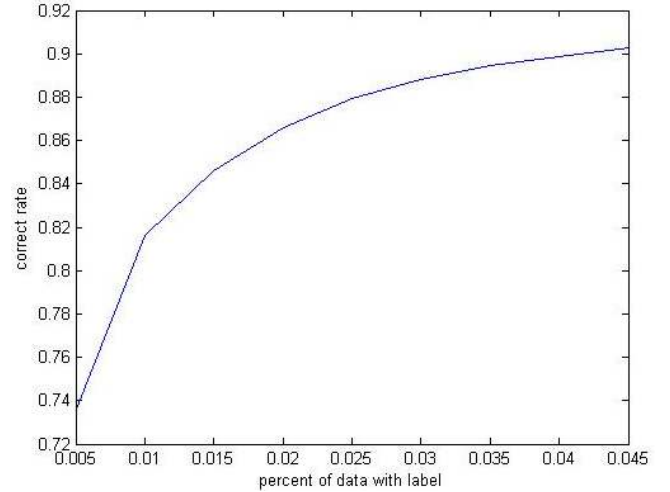


FIGURE 5 – MNIST semi-supervised right detection

4.2 Semi-supervised and unsupervised learning

Now, we will try to see how effective the semi-supervision is with the MNIST dataset. We will use 0.5% up to 4.5% of the dataset (i.e. 300 to 2700 pictures) alongside the k-NN algorithm. Then, we see how the unused data are classified according to this small sample. Even with only 300 pictures, the classification is right in approximately 73% ; with 2700 pictures, it goes up just above 90%, making semi-supervised learning rather adapted for MNIST classification (figure 5).

Then, we tried to see how the k-NN algorithm performs after a dimension reduction. We applied the t-SNE algorithm on a small sample of the MNIST dataset (2400 pictures). Then, we transformed the unlabeled data through the t-SNE used for the labeled data and finally, we determined classes with the k-NN algorithm (figure 6). Compared to the original labels (figure 7), the results are deceiving with only 17% labels matching.

5 EEG results

5.1 Data processing

The EEG dataset contains seven different labels : six are for the actions that were asked to

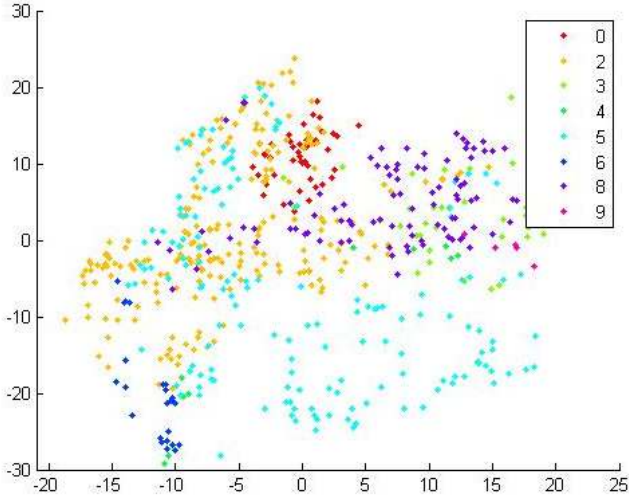


FIGURE 6 – MNIST repartition with the new labels

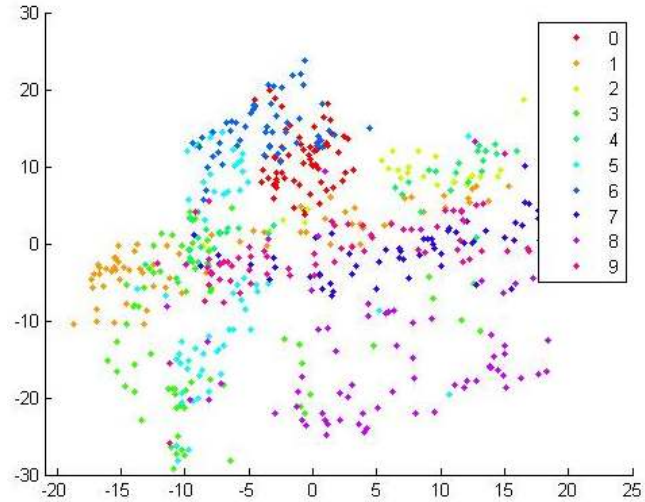


FIGURE 7 – MNIST repartition of known labels

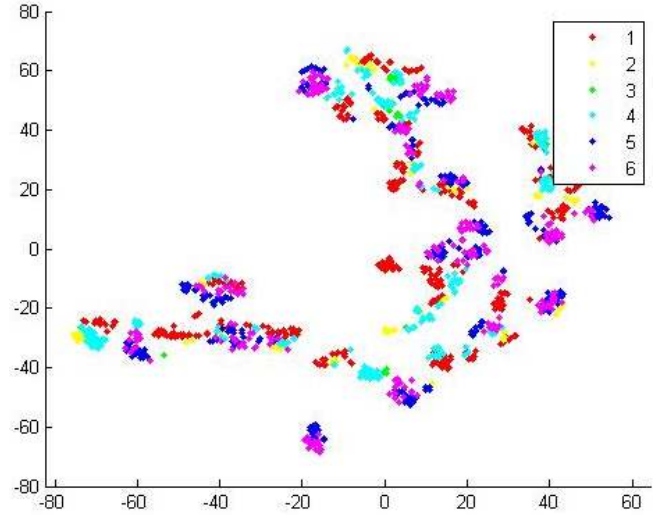


FIGURE 8 – 2D repartition of the EEG data after t-SNE

perform :

1. HandStart
2. FirstDigitTouch
3. BothStartLoadPhase
4. LiftOff
5. Replace
6. BothReleased

0. Every other action or transition moment

The dataset contained two major issues : the sheer number of 0 (they constituted almost 90% of the dataset) and the high variability of the successive moments of a unique action. We chose to remove the graphs labeled under 0 and we averaged every successive moments of a same action into one. This results in a dataset of 17814 graphs.

5.2 Visualization

First, we begin by using the t-SNE algorithm on the EEG dataset ; however neither 2D nor 3D reduction has proven effective in determining different groups (figure 8) (figure 9). A One-versus-the-rest shows that even a single data is mixed with the others elements - no label present a clear distinction (figure 10).

5.3 Semi-supervised learning

We tried to do supervised learning on a random 2100 subset of the EEG dataset for data

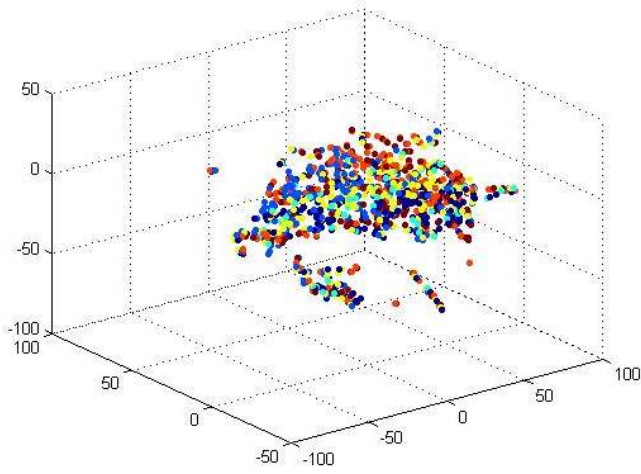


FIGURE 9 – 3D repartition of the EEG data after t-SNE

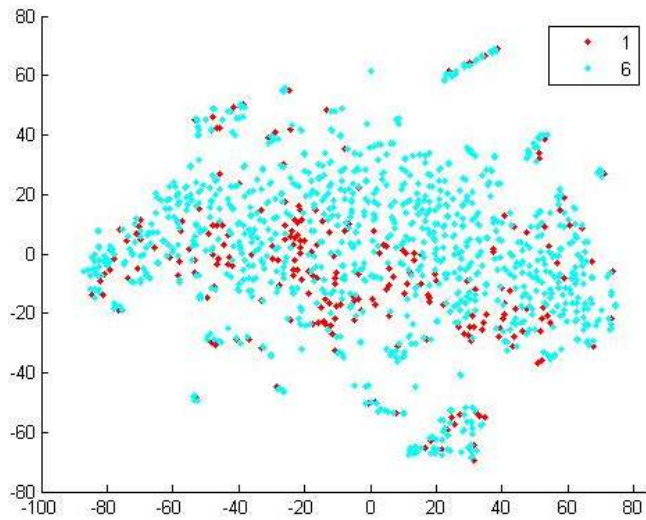


FIGURE 10 – Exemple of 1 versus 2/3/4/5/6 in EEG

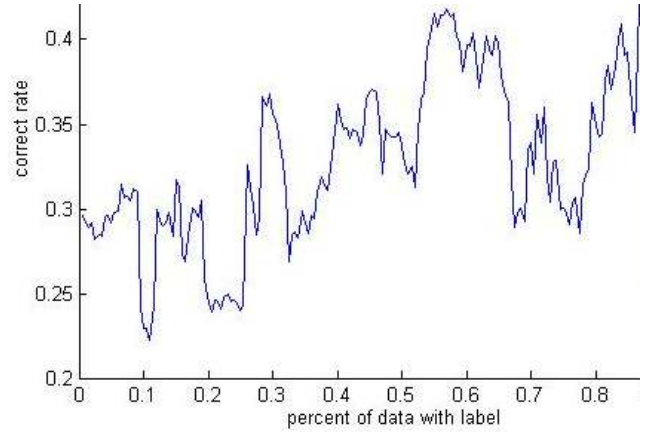


FIGURE 11 – EEG semi-supervised right detection

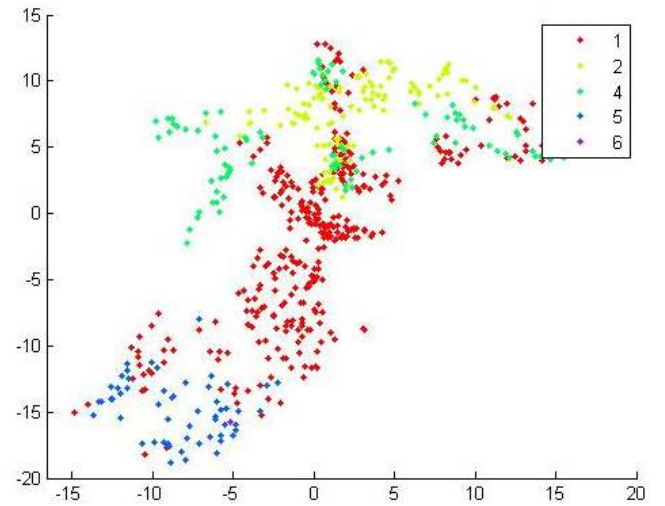


FIGURE 12 – EEG repartition with the new labels

size limitation. The result of a single test is shown in (figure 9). The correct rate seems to be growing with the size of the sample with labels ; furthermore, the rate of correct guesses seems to be above 23% in the worst case, making it a little better than a random classification. The k-NN is not really effective here, but considering the scattering of the points in 2D and 3D, the k-NN isn't so bad either, with a maximum correct rate of 40% in the sample.

Then, we did as with the MNIST a k-NN learning after a t-SNE on a fraction of the dataset (figure 11) (figure 12). The rate of correct guesses drop to 30,5%.

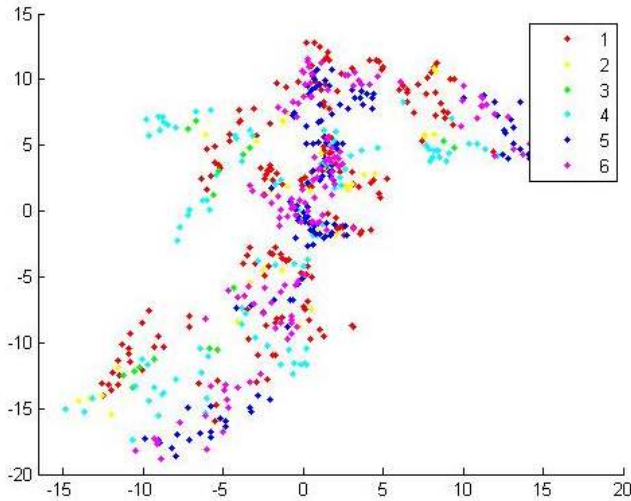


FIGURE 13 – EEG repartition of known labels

6 Results comparison

We can see that the MNIST dataset functions relatively well with the t-SNE and k-NN algorithm, compared to the EEG dataset that had some issues notably with t-SNE. However, the semi-supervised learning performed relatively well with the EEG dataset, especially considering the previous results.

One very limiting factor was the maximum amount of data we were able to use due to MatLab limitations; a second series of experimentations with a language supporting larger amount of data would be an improvement. For the EEG visualization, reducing the data on a single individual could improve the effectiveness, but a few one-versus-the-rest that has been made accordingly did not seem to really improve on the original dataset.

Graph visualization through t-SNE has proven to be a good way to see if a dataset has well-defined classes; furthermore, the semi-supervised learning after a dimension reduction does not seem that bad considering the 30% of correct guesses with the EEG dataset. However, this number needs to be confronted to the 17% of the MNIST; this may be either the result of an error in the way we proceed or just a coincidence. In all cases, this should be further investigated.

7 Conclusion

Dimension reduction is a powerful tool that can have applications not only in visualization, but in machine learning too. However, its effectiveness is very dependent of the quality of the dataset and it will probably perform best on well-separated datasets. Finally, the impact of dimension reduction on semi-supervised learning seems to be relatively negative, but we cannot say for sure how strong is the impact.

8 References

- [1] <https://www.kaggle.com/c/digit-recognizer/data> (last consulted the 12/05/2016)
- [2] <https://www.kaggle.com/c/grasp-and-lift-ecg-detection/data> (last consulted the 12/05/2016)
- [3] L.J.P. van der Maaten, G.E. Hinton, Visualizing High-Dimensional Data Using t-SNE, 2008.